

ASSENS TOROE, Denmark

WMO: 061230

Lat: 55.2444N

Lon: 9.8883E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.3	-3.9	-8.6	1.8	-4.1	-7.1	2.1	-2.8	16.0	4.7	14.6	5.0	4.5	70	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	25.4	19.4	23.8	18.6	22.2	17.9	20.3	24.0	19.3	22.6	18.4	21.4	4.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	13.7	22.5	18.0	12.9	21.3	17.1	12.2	20.2	58.3	24.1	55.0	22.7	52.2	21.4	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	12.3	11.0	-6.6	28.4	2.4	1.6	-8.3	29.6	-9.7	30.5	-11.1	31.4	-12.9	32.6		
(5)			-7.2	21.6	2.3	1.1	-8.9	22.4	-10.2	23.0	-11.5	23.7	-13.2	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	1.9	1.8	3.9	8.1	12.4	15.4	18.0	18.0	15.2	10.8	6.9	3.9
	DBStd	6.44	3.29	2.85	2.90	2.81	2.96	2.38	2.62	2.11	2.17	2.64	2.73	3.19
	HDD10.0	1054	250	229	189	71	9	0	0	0	0	22	97	188
	HDD18.3	3193	508	462	448	306	186	93	40	31	98	232	343	446
	CDD10.0	963	0	0	0	14	83	162	247	248	155	48	4	0
	CDD18.3	58	0	0	0	0	1	4	29	21	3	0	0	0
	CDH23.3	181	0	0	0	0	4	12	112	51	3	0	0	0
	CDH26.7	16	0	0	0	0	0	0	12	4	0	0	0	0
Wind	WSAvg	5.7	6.5	5.9	5.4	4.9	5.0	5.4	5.2	5.3	5.7	6.1	6.5	6.6
Precipitation	PrecAvg	739	70	50	45	35	50	63	68	80	70	74	65	69
	PrecMax	940	147	122	106	89	84	143	149	158	147	175	138	145
	PrecMin	509	3	8	6	2	14	1	13	28	12	28	15	18
	PrecStd	116	35	25	25	19	22	31	36	37	36	28	31	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	8.3	12.8	18.3	23.6	25.1	28.2	27.1	23.3	17.3	13.6	10.1
		MCWB	8.0	6.8	8.9	12.2	17.3	18.3	20.1	20.1	18.1	15.0	12.5	9.2
	2%	DB	8.0	7.1	9.8	16.0	21.0	22.7	26.1	24.7	21.1	15.9	12.4	9.2
		MCWB	7.1	5.9	7.1	11.0	15.1	17.3	19.6	19.4	17.2	14.0	11.3	8.3
	5%	DB	7.2	6.3	8.3	14.2	19.0	21.1	24.4	23.1	19.6	15.1	11.4	8.6
		MCWB	6.3	5.2	6.3	10.5	14.6	16.9	19.1	18.5	16.3	13.5	10.3	7.7
	10%	DB	6.4	5.5	7.2	12.4	17.2	19.5	22.6	21.5	18.4	14.3	10.5	7.8
		MCWB	5.5	4.6	5.7	9.3	13.6	16.0	18.5	17.8	15.7	12.9	9.5	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	7.2	9.1	13.1	18.4	19.7	21.9	21.2	19.0	15.6	12.8	9.5
		MCDB	8.8	7.8	12.1	16.8	22.7	22.8	26.1	25.2	21.9	16.8	13.5	9.9
	2%	WB	7.2	6.3	7.6	11.8	16.4	18.5	20.7	20.0	17.7	14.5	11.5	8.5
		MCDB	7.8	6.9	9.5	14.9	19.6	21.5	24.5	23.8	20.1	15.5	12.2	9.0
	5%	WB	6.4	5.4	6.7	10.7	15.1	17.4	19.7	19.1	16.9	13.8	10.5	7.9
		MCDB	7.1	6.1	7.8	13.4	18.0	20.0	23.1	22.1	19.0	14.7	11.2	8.5
	10%	WB	5.5	4.7	6.0	9.7	13.9	16.4	18.7	18.2	16.2	13.1	9.6	7.1
		MCDB	6.3	5.4	7.0	12.0	16.8	18.9	22.0	20.9	18.1	14.1	10.4	7.8
Mean Daily Temperature Range	5% DB	MDBR	2.8	3.0	4.3	5.9	6.2	6.1	6.0	5.2	4.6	3.5	2.8	2.8
		MCDBR	2.9	3.7	6.0	8.7	9.1	8.7	8.9	7.9	6.3	4.0	2.6	2.6
		MCWBR	3.0	3.0	4.0	5.1	5.4	5.3	4.5	4.1	3.8	3.0	2.6	2.6
	5% WB	MCDBR	3.0	3.5	5.2	7.8	8.3	7.6	8.0	6.9	5.6	3.6	2.6	2.6
		MCWBR	3.1	3.0	3.7	4.9	5.4	5.2	4.7	4.2	3.8	3.0	2.7	2.7
Clear-Sky Solar Irradiance	taub	0.320	0.340	0.366	0.399	0.392	0.387	0.404	0.404	0.381	0.358	0.337	0.307	
	taud	2.268	2.295	2.270	2.229	2.320	2.363	2.331	2.344	2.390	2.399	2.329	2.255	
	Ebn at Noon	592	726	795	819	851	859	833	809	777	700	570	515	
	Edh at Noon	57	79	102	122	118	114	116	108	91	71	54	47	
All-Sky Solar Radiation	RadAvg	0.48	1.10	2.45	4.10	5.28	5.59	5.22	4.18	2.94	1.52	0.61	0.35	
	RadStd	0.08	0.16	0.23	0.48	0.52	0.42	0.57	0.30	0.30	0.18	0.09	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	-154	N/A	N/A

Nomenclature: See separate page